

Tomato Leaf Disease Guide

Septoria Leaf Spot · Bacterial Leaf Spot · Early Blight · Late Blight

Before you use this guide: This covers general, widely-recommended practices. Exact fungicide/bactericide products and dosages vary by state and change as regulations update — always confirm the specific product and dose with your nearest Krishi Vigyan Kendra (KVK) or state agricultural extension officer, and follow the product label exactly. This guide prepares you for that conversation; it doesn't replace it.

1. Septoria Leaf Spot

What it looks like

Small, circular brown-to-tan spots on the leaves, usually starting on the lower/older leaves near the ground. Look closely at the center of a spot — Septoria often shows tiny black dots (like pepper flecks) in the middle, the clearest way to tell it apart from early blight. It does not infect the fruit or stem, only the leaves.

Conditions that favor it

Moderate temperatures (20-25°C) combined with long, continuous periods of leaf wetness. A single wet day is a weak signal; several wet days in a row is when this disease actually takes hold.

What to do

- **Irrigation:** Switch to drip irrigation or water at the base of the plant. If overhead watering is unavoidable, water early morning so leaves dry during the day.
- **Spacing & mulch:** Space plants well apart and mulch the soil to stop rain from splashing soil (where the fungus survives) onto leaves.
- **Sanitation:** Remove infected lower leaves as soon as seen. Clear all old plant debris at season's end.
- **Treatment:** Copper-based fungicides are commonly effective when applied early. Confirm the current approved product/dose with your local KVK.

2. Early Blight

What it looks like

Circular brown spots, usually larger than Septoria's, with a distinctive target pattern — rings inside rings — often surrounded by a yellow ring. Starts on the oldest, lowest leaves and moves upward; can later affect stems and fruit.

Conditions that favor it

Warm temperatures (24-29°C) with high humidity, especially when wet and dry periods alternate. Stressed plants (nutrient-deficient, drought-stressed, fruit-overloaded) are more vulnerable.

What to do

- **Irrigation:** Drip over overhead watering; never water late in the day.
- **Soil & nutrition:** Avoid excess nitrogen. Use well-decomposed farmyard manure or compost, not fresh manure.
- **Crop rotation:** Avoid tomato, potato, brinjal, or chili in the same field for at least 2 years.
- **Pruning & staking:** Stake plants and prune lowest leaves for airflow.

- **Treatment:** Chlorothalonil or mancozeb fungicides are commonly used preventatively; copper-based products also help. Spraying an already-infected leaf won't cure it — the goal is protecting healthy growth.

3. Bacterial Leaf Spot

What it looks like

Small spots (much smaller than early blight), often water-soaked or greasy-looking, especially on the leaf underside. Centers can dry and fall away, leaving a "shot-hole" look. Unlike Septoria, this disease also affects the fruit.

Conditions that favor it

Warm temperatures (24-30°C) combined with actual rain or wind-driven splash events — needs a physical splash/wounding event to spread.

What to do

This is the hardest of the four to treat once established — prevention matters more here than for any other disease on this list, since no fully resistant commercial variety exists and there is no true cure once infected.

- **Irrigation:** Avoid overhead irrigation entirely if possible — the single most important step for this disease.
- **Seed source:** Use certified, disease-free seed/seedlings from a reputable nursery.
- **Hygiene:** Don't handle plants when foliage is wet. Disinfect tools and stakes between uses.
- **Treatment:** Copper-based bactericide sprays are the main option, and work best as prevention rather than cure. Check current local guidance before use.

4. Late Blight

What it looks like

The most destructive and fastest-spreading of the four. Pale green-to-brown, water-soaked patches with a light green-to-yellow halo, appearing anywhere on the plant, not just lower leaves. White fuzzy fungal growth may appear on the underside of leaves in humid conditions. Can destroy a field within days.

Conditions that favor it

Cool temperatures (15-22°C) with high humidity (above 85-90%) and extended wetness — the opposite temperature pattern from early blight and bacterial spot.

What to do

- **Treat any suspected case as an emergency** — report to your local agricultural extension office immediately given how fast and contagiously it spreads between neighboring farms.
- **Irrigation:** Drip irrigation; avoid overhead watering, especially in cool weather.
- **Remove infected material immediately** — pull and destroy (don't compost) severely infected plants.
- **Resistant varieties:** Ask your local KVK which late-blight-tolerant varieties suit your region and season.
- **Treatment:** Preventive fungicide programs (applied before symptoms, during known high-risk weather) are far more effective than treating after an outbreak starts. Ask your extension officer for currently recommended products.

General Practices That Help Against All Four Diseases

Irrigation

- Drip irrigation (or watering at the base) beats overhead sprinkler watering for every disease on this list — wet leaves are the common thread.
- If overhead watering is your only option, always water early morning so leaves dry during the day.
- Avoid evening watering — leaves that stay wet overnight create exactly the conditions these diseases need.

Soil health & manure

- Use well-decomposed farmyard manure (FYM) or compost, not fresh/raw manure, which can carry pathogens and release nitrogen too quickly.
- Avoid over-fertilizing with nitrogen; adequate potassium supports the plant's own disease resistance.
- Vermicompost and neem cake are common Indian soil-health inputs that support overall plant vigor alongside these practices, not instead of them.

Field hygiene

- Rotate tomatoes and related crops (potato, brinjal, chili) out of a field for at least 2 years if any of these diseases have appeared before.
- Remove and destroy infected material — don't leave it in the field or add heavily diseased material to compost.
- Space plants well and prune lower leaves to keep air moving through the canopy.

When in doubt

These four diseases can look similar in early stages. If you're not sure which one you're looking at, use the CropGuard scan feature for a second opinion, and bring a sample to your local KVK for confirmation before spraying anything.

This guide was compiled from general recommendations published by university agricultural extension services and is intended as an educational starting point for farmers. It is not a substitute for advice from a qualified local agronomist or your state agricultural extension office, particularly regarding specific product choice, dosage, and regional regulations.